

SAR Platform API Documentation

1. Thông tin chung

– Base URL mặc định:

○ HTTPS: <https://sar-be.onrender.com>

– Prefix API: phần lớn endpoint bắt đầu bằng /api.

– Content-Type cho body JSON: [application/json](#).

– Auth: các endpoint có [Authorize] cần header Authorization: Bearer {accessToken}.

– Permission: nhiều endpoint còn cần JWT claim permission khớp với permission code ghi trong tài liệu.

– Tài liệu OpenAPI/Scalar chỉ được map ở môi trường Development.

- Admin account:

username: admin

password: Admin@123

Mẫu lỗi chuẩn

Global exception middleware trả về JSON dạng:

```
{
  "statusCode": 400,
  "message": "One or more validation errors occurred.",
  "traceId": "0HN...",
  "errors": ["Email is invalid."]
}
```

Mapping chính:

Exception	HTTP	Ghi chú
ValidationException	400	Lỗi validate input/nghiệp vụ
DomainException	400	Lỗi domain
UnauthorizedException	401	Sai token/login/refresh token

NotFoundException	404	Không tìm thấy resource
ConflictException	409	Dữ liệu bị trùng/xung đột
MetaGraphApiException	502	Lỗi khi gọi Meta Graph API
Lỗi khác	500	Production trả message chung

Lưu ý: nếu request bị chặn bởi auth/permission middleware, response 401 hoặc 403 có thể không có JSON body.

Header dùng chung

Authorization: Bearer eyJ...

Content-Type: application/json

2. Auth và tài khoản hiện tại

2.1 Login

- Khi FE gọi: màn hình đăng nhập.
- Endpoint: **POST** /api/auth/login
- Auth: không cần.

Request:

```
{
  "usernameOrEmail": "admin@sar.local",
  "password": "P@ssw0rd!"
}
```

Response **200**:

```
{
  "accessToken": "eyJ...",
  "refreshToken": "a-refresh-token",
  "tokenType": "Bearer",
  "expiresIn": 3600,
  "user": {
    "id": "11111111-1111-1111-1111-111111111111",
    "username": "admin",
    "email": "admin@sar.local",
    "firstTimeLogin": false
  }
},
```

```
"role": {
  "id": "22222222-2222-2222-2222-222222222222",
  "name": "Admin",
  "code": "ADMIN"
},
"permissions": ["ROLE_MANAGEMENT.ROLE_LIST.VIEW"]
}
```

Lỗi thường gặp:

– 400: UsernameOrEmail is required., Password is required.

– 401: Invalid username/email or password.

FE note: lưu accessToken, refreshToken, permission list và role. Dùng permission list để ẩn/hiện menu/action.

2.2 Refresh token

– Khi FE gọi: access token hết hạn hoặc sắp hết hạn.

– Endpoint: **POST** /api/auth/refresh-token

– Auth: không cần header Bearer, nhưng body cần token cũ.

Request:

```
{
  "accessToken": "old-access-token",
  "refreshToken": "stored-refresh-token"
}
```

Response **200**:

```
{
  "accessToken": "new-access-token",
  "refreshToken": "new-refresh-token",
  "tokenType": "Bearer",
  "expiresIn": 3600,
  "permissions": ["SOCIAL_MEDIA.FACEBOOK_INTEGRATION.VIEW"]
}
```

Lỗi thường gặp:

– 401: Invalid token., Invalid refresh token.

– 401: Role is inactive.

FE note: refresh token có thể rotate, luôn thay cả access token và refresh token bằng giá trị mới.

2.3 Logout

- Khi FE gọi: user đăng xuất.
- Endpoint: **POST /api/auth/logout**
- Auth: không bắt buộc trong controller hiện tại.

Request:

```
{
  "refreshToken": "stored-refresh-token"
}
```

Response **200**:

```
{
  "success": true
}
```

2.4 Lấy profile user hiện tại

- Khi FE gọi: sau login, reload app, màn hình profile.
- Endpoint: **GET /api/me**
- Auth: Bearer token.

Response **200**:

```
{
  "id": "11111111-1111-1111-1111-111111111111",
  "username": "admin",
  "email": "admin@sar.local",
  "firstTimeLogin": false,
  "role": {
    "id": "22222222-2222-2222-2222-222222222222",
    "name": "Admin",
    "code": "ADMIN"
  }
}
```

2.5 Lấy permissions và menu của user hiện tại

- Khi FE gọi: khởi tạo navigation/menu và action guards.
- Endpoint: **GET /api/me/permissions**
- Auth: Bearer token.

Response 200:

```
{
  "user": {
    "id": "11111111-1111-1111-1111-111111111111",
    "username": "admin",
    "email": "admin@sar.local"
  },
  "role": {
    "id": "22222222-2222-2222-2222-222222222222",
    "name": "Admin",
    "code": "ADMIN"
  },
  "permissions": ["PERMISSION_MANAGEMENT.FEATURE_CONFIG.VIEW"],
  "menus": [
    {
      "id": "33333333-3333-3333-3333-333333333333",
      "name": "Permission Management",
      "code": "PERMISSION_MANAGEMENT",
      "displayOrder": 1,
      "pages": [
        {
          "id": "44444444-4444-4444-4444-444444444444",
          "name": "Feature Config",
          "code": "FEATURE_CONFIG",
          "route": "/permissions/features",
          "icon": "settings",
          "displayOrder": 1
        }
      ]
    }
  ]
}
```

3. Business Partners

Tất cả API trong mục này cần Bearer token.

3.1 Danh sách business partners

- Khi FE gọi: màn hình list/search/filter.
- Endpoint: **GET**
/api/business-partners?keyword=&isActive=&pageNumber=1&pageSize=20
- Permission: **BUSINESS_PARTNER.PROFILE.VIEW**
- Query:

- **keyword**: optional.
- **isActive**: optional boolean.
- **pageNumber**: mặc định 1.
- **pageSize**: mặc định 20, service giới hạn tối đa 100.

Response **200**:

```
{
  "items": [
    {
      "id": "aaaaaaaa-aaaa-aaaa-aaaa-aaaaaaaaaaaa",
      "brandName": "SAR Coffee",
      "logoUrl": "https://cdn.example.com/logo.png",
      "email": "owner@sarcoffee.vn",
      "phone": "0900000000",
      "representativeName": "Nguyễn Văn A",
      "representativeEmail": "rep@sarcoffee.vn",
      "isActive": true,
      "createdAt": "2026-06-16T10:00:00Z",
      "updatedAt": null,
      "accountCreated": null,
      "representativeEmailSent": null,
      "businessOwnerEmailSent": null
    }
  ],
  "pageNumber": 1,
  "pageSize": 20,
  "totalCount": 1
}
```

3.2 Chi tiết business partner

- Khi FE gọi: trang detail/edit.
- Endpoint: **GET /api/business-partners/{businessPartnerId}**
- Permission: **BUSINESS_PARTNER.PROFILE.VIEW**

Response **200**:

```
{
  "id": "aaaaaaaa-aaaa-aaaa-aaaa-aaaaaaaaaaaa",
  "brandName": "SAR Coffee",
  "logoUrl": null,
  "email": "owner@sarcoffee.vn",
  "phone": "0900000000",
  "representativeName": "Nguyễn Văn A",
  "representativeEmail": "rep@sarcoffee.vn",
  "isActive": true,
}
```

```
"createdAt": "2026-06-16T10:00:00Z",
"updatedAt": null,
"usersCount": 1,
"integrationsCount": 0
}
```

Lỗi: 404 Business partner was not found.

3.3 Tạo business partner

- Khi FE gọi: submit form tạo business partner.
- Endpoint: **POST /api/business-partners**
- Permission: **BUSINESS_PARTNER.PROFILE.CREATE**
- Status thành công: 201 Created.

Request:

```
{
  "brandName": "SAR Coffee",
  "logoUrl": "https://cdn.example.com/logo.png",
  "email": "owner@sarcoffee.vn",
  "phone": "0900000000",
  "representativeName": "Nguyễn Văn A",
  "representativeEmail": "rep@sarcoffee.vn"
}
```

Response 201:

```
{
  "id": "aaaaaaaa-aaaa-aaaa-aaaa-aaaaaaaaaaaa",
  "brandName": "SAR Coffee",
  "logoUrl": "https://cdn.example.com/logo.png",
  "email": "owner@sarcoffee.vn",
  "phone": "0900000000",
  "representativeName": "Nguyễn Văn A",
  "representativeEmail": "rep@sarcoffee.vn",
  "isActive": true,
  "createdAt": "2026-06-16T10:00:00Z",
  "updatedAt": null,
  "accountCreated": true,
  "representativeEmailSent": true,
  "businessOwnerEmailSent": true
}
```

Lỗi thường gặp:

- 400: thiếu/sai BrandName, Email, Phone, RepresentativeName, RepresentativeEmail.
- 409: Business partner email is already used.

- 409: Representative email is already used by an existing user.
- 400: role representative thiếu/inactive.

FE note quan trọng: BE tự tạo account representative và gửi email. FE không cần và không được hiện password vì API không return password. Nên hiện thông báo dựa trên 3 flag accountCreated, representativeEmailSent, businessOwnerEmailSent.

3.4 Cập nhật business partner

- Endpoint: **PUT /api/business-partners/{businessPartnerId}**
- Permission: **BUSINESS_PARTNER.PROFILE.UPDATE**

Request:

```
{
  "brandName": "SAR Coffee Updated",
  "logoUrl": null,
  "email": "owner@sarcoffee.vn",
  "phone": "0911111111",
  "representativeName": "Nguyễn Văn B",
  "representativeEmail": "rep@sarcoffee.vn",
  "isActive": true
}
```

Response **200**: **BusinessPartnerResponse**.

Lỗi: tương tự create, thêm **404 Business partner was not found**.

3.5 Xóa business partner

- Endpoint: **DELETE /api/business-partners/{businessPartnerId}**
- Permission: **BUSINESS_PARTNER.PROFILE.DELETE**
- Response: **204 No Content**.

Lỗi: **404 Business partner was not found**.

4. Permission, role và menu configuration

Tất cả API trong mục này cần Bearer token và permission tương ứng. Các resource system-defined có thể bị bảo vệ bởi service, tùy thông điệp lỗi trả về từ source hiện tại.

4.1 Roles

API	Permission	Khi FE gọi
-----	------------	------------

GET /api/roles?keyword=&isActive=&pageNumber=1&pageSize=20	ROLE_MANAGEMENT.ROLE_LIST.VIEW	List role
GET /api/roles/{roleId}	ROLE_MANAGEMENT.ROLE_LIST.VIEW	Detail/edit role
POST /api/roles	ROLE_MANAGEMENT.ROLE_LIST.CREATE	Tạo role
PUT /api/roles/{roleId}	ROLE_MANAGEMENT.ROLE_LIST.UPDATE	Cập nhật role
DELETE /api/roles/{roleId}	ROLE_MANAGEMENT.ROLE_LIST.DELETE	Xóa role
GET /api/roles/{roleId}/permissions/matrix	PERMISSION_MANAGEMENT.ROLE_PERMISSION_CONFIG.VIEW	Màn hình gán quyền
PUT /api/roles/{roleId}/permissions	PERMISSION_MANAGEMENT.ROLE_PERMISSION_CONFIG.UPDATE	Replace toàn bộ quyền role
PATCH /api/roles/{roleId}/permissions	PERMISSION_MANAGEMENT.ROLE_PERMISSION_CONFIG.UPDATE	Grant/revoke một phần
POST /api/roles/{targetRoleId}/permissions/copy-from/{sourceRoleId}	PERMISSION_MANAGEMENT.ROLE_PERMISSION_CONFIG.UPDATE	Copy quyền từ role khác

Create role request:

```
{
  "name": "Business Operator",
  "code": "BUSINESS_OPERATOR",
  "description": "Operate business workflows"
}
```

Role response:

```
{
  "id": "bbbbbbbbb-bbbb-bbbb-bbbb-bbbbbbbbbbbbb",
  "name": "Business Operator",
  "code": "BUSINESS_OPERATOR",
  "description": "Operate business workflows",
  "isSystemRole": false,
  "isActive": true,
  "createdAt": "2026-06-16T10:00:00Z",
  "updatedAt": null
}
```

Replace permissions request:

```
{
  "pageActionIds": [
    "cccccccc-cccc-cccc-cccc-cccccccccccc"
  ]
}
```

Patch permissions request:

```
{
  "grantPageActionIds": ["cccccccc-cccc-cccc-cccc-cccccccccccc"],
  "revokePageActionIds": ["dddddddd-dddd-dddd-dddd-dddddddddddd"]
}
```

Permission update response:

```
{
  "roleId": "bbbbbbbbb-bbbb-bbbb-bbbb-bbbbbbbbbbbbb",
  "totalGrantedPermissions": 1,
  "grantedPermissionCodes": ["SOCIAL_MEDIA.FACEBOOK_INTEGRATION.VIEW"]
}
```

Lỗi thường gặp: Role not found., Role code already exists., Cannot update permissions for inactive role., All pageActionIds must exist and be active., duplicate/empty ids.

4.2 Features

API	Permission
GET /api/features?keyword=&isActive=	PERMISSION_MANAGEMENT.FEATURE_CONFIG.VIEW
GET /api/features/{featureId}?includePages=true	PERMISSION_MANAGEMENT.FEATURE_CONFIG.VIEW

POST /api/features	PERMISSION_MANAGEMENT.FEATURE_CONFIG. CREATE
PUT /api/features/{featureId}	PERMISSION_MANAGEMENT.FEATURE_CONFIG. UPDATE
DELETE /api/features/{featureId}	PERMISSION_MANAGEMENT.FEATURE_CONFIG. DELETE

Create request:

```
{
  "systemModuleDefinitionId": "eeeeeeee-eeee-eeee-eeee-eeeeeeeeeeee",
  "name": "Social Media",
  "description": "Social media integration",
  "displayOrder": 10
}
```

Response:

```
{
  "id": "ffffff-ffff-ffff-ffff-ffffff",
  "systemModuleDefinitionId": "eeeeeeee-eeee-eeee-eeee-eeeeeeeeeeee",
  "name": "Social Media",
  "code": "SOCIAL_MEDIA",
  "description": "Social media integration",
  "displayOrder": 10,
  "isSystemDefined": true,
  "isActive": true,
  "pages": null
}
```

Lỗi: Feature not found., SystemModuleDefinition not found., SystemModuleDefinition is inactive., System feature code already exists.

4.3 Pages

API	Permission
GET /api/pages?featureId=&keyword=&isActive=&isMenuVisible=	PERMISSION_MANAGEMENT.PAGE_CONFIG.VIEW
GET /api/pages/{pageId}?includeActions=true	PERMISSION_MANAGEMENT.PAGE_CONFIG.VIEW

POST /api/pages	PERMISSION_MANAGEMENT.PAGE_CONFIG.CREATE
PUT /api/pages/{pageId}	PERMISSION_MANAGEMENT.PAGE_CONFIG.UPDATE
DELETE /api/pages/{pageId}	PERMISSION_MANAGEMENT.PAGE_CONFIG.DELETE

Create request:

```
{
  "featureId": "ffffffff-ffff-ffff-ffffffffffff",
  "systemPageDefinitionId": "12121212-1212-1212-1212-121212121212",
  "name": "Facebook Integration",
  "route": "/social/facebook",
  "icon": "facebook",
  "displayOrder": 1,
  "isMenuVisible": true
}
```

Response:

```
{
  "id": "13131313-1313-1313-1313-131313131313",
  "featureId": "ffffffff-ffff-ffff-ffffffffffff",
  "featureName": "Social Media",
  "featureCode": "SOCIAL_MEDIA",
  "featureIsSystemDefined": true,
  "systemPageDefinitionId": "12121212-1212-1212-1212-121212121212",
  "name": "Facebook Integration",
  "code": "FACEBOOK_INTEGRATION",
  "route": "/social/facebook",
  "icon": "facebook",
  "displayOrder": 1,
  "isMenuVisible": true,
  "isSystemDefined": true,
  "isActive": true,
  "actions": null
}
```

Lỗi: Page not found., SystemPageDefinition not found., SystemPageDefinition is inactive., SystemPageDefinition does not belong to the feature module definition., System page code already exists in this feature.

4.4 Permission actions

API	Permission
GET /api/permission-actions?keyword=&isActive= =	PERMISSION_MANAGEMENT.ACTION_CONFIG. VIEW
POST /api/permission-actions	PERMISSION_MANAGEMENT.ACTION_CONFIG. CREATE
PUT /api/permission-actions/{permissionActionId}	PERMISSION_MANAGEMENT.ACTION_CONFIG. UPDATE
DELETE /api/permission-actions/{permissionActionId}	PERMISSION_MANAGEMENT.ACTION_CONFIG. DELETE

Create request:

```
{
  "systemActionDefinitionId": "14141414-1414-1414-1414-141414141414",
  "name": "View",
  "description": "Can view page"
}
```

Response:

```
{
  "id": "15151515-1515-1515-1515-151515151515",
  "systemActionDefinitionId": "14141414-1414-1414-1414-141414141414",
  "name": "View",
  "code": "VIEW",
  "description": "Can view page",
  "isSystemDefined": true,
  "isActive": true
}
```

Lỗi: PermissionAction not found., SystemActionDefinition not found., SystemActionDefinition is inactive., System action code already exists.

4.5 Page actions

API	Permission
-----	------------

GET /api/page-actions?featureId=&pageId=&permissionActionId=&keyword=&isActive=	PERMISSION_MANAGEMENT.PAGE_ACTION_CONFIG.VIEW
POST /api/page-actions	PERMISSION_MANAGEMENT.PAGE_ACTION_CONFIG.CREATE
PUT /api/page-actions/{pageActionId}	PERMISSION_MANAGEMENT.PAGE_ACTION_CONFIG.UPDATE
DELETE /api/page-actions/{pageActionId}	PERMISSION_MANAGEMENT.PAGE_ACTION_CONFIG.DELETE

Create request:

```
{
  "pageId": "13131313-1313-1313-1313-131313131313",
  "permissionActionId": "15151515-1515-1515-1515-151515151515",
  "description": "Can view Facebook integration"
}
```

Response:

```
{
  "id": "16161616-1616-1616-1616-161616161616",
  "featureId": "ffffffff-ffff-ffff-ffffffffffffff",
  "featureName": "Social Media",
  "featureCode": "SOCIAL_MEDIA",
  "pageId": "13131313-1313-1313-1313-131313131313",
  "pageName": "Facebook Integration",
  "pageCode": "FACEBOOK_INTEGRATION",
  "permissionActionId": "15151515-1515-1515-1515-151515151515",
  "actionName": "View",
  "actionCode": "VIEW",
  "permissionCode": "SOCIAL_MEDIA.FACEBOOK_INTEGRATION.VIEW",
  "description": "Can view Facebook integration",
  "isSystemDefined": true,
  "isActive": true
}
```

Lỗi: PageAction already exists for this page and action., PermissionCode already exists., PageAction not found., Page is inactive., PermissionAction is inactive.

4.6 System permission definitions

API	Permission	Mục đích
-----	------------	----------

GET /api/system-module-definitions	PERMISSION_MANAGEMENT.FEATURE_CONFIG.VIEW	Lấy module seed/system để tạo feature
GET /api/system-module-definitions/{id}/page-definitions	PERMISSION_MANAGEMENT.PAGE_CONFIG.VIEW	Lấy page definitions theo module
GET /api/system-action-definitions	PERMISSION_MANAGEMENT.ACTION_CONFIG.VIEW	Lấy action definitions

System definition response mẫu:

```
{
  "id": "eeeeeeee-eeee-eeee-eeee-eeeeeeeeeeee",
  "name": "Social Media",
  "code": "SOCIAL_MEDIA",
  "description": "Social media module",
  "isActive": true
}
```

5. Social media integration theo Business Partner

Đây là flow quản trị Facebook integration được gắn với businessPartnerId, có lưu app config/app secret và page access token trong backend.

5.1 Danh sách social media providers

- Endpoint: GET /api/social-media/providers
- Permission: SOCIAL_MEDIA.FACEBOOK_INTEGRATION.VIEW

Response:

```
[
  {
    "id": "17171717-1717-1717-1717-171717171717",
    "name": "Facebook",
    "code": "FACEBOOK",
    "description": "Facebook Messenger",
    "isActive": true
  }
]
```

5.2 Danh sách integrations của business partner

- Endpoint: `GET /api/business-partners/{businessPartnerId}/social-media/integrations`
- Permission: `SOCIAL_MEDIA.FACEBOOK_INTEGRATION.VIEW`

Response:

```
[
  {
    "id": "18181818-1818-1818-1818-181818181818",
    "providerName": "Facebook",
    "providerCode": "FACEBOOK",
    "appld": "123456789",
    "status": "Authorized",
    "authorizedAt": "2026-06-16T10:00:00Z",
    "pagesCount": 2,
    "activeBotPagesCount": 1
  }
]
```

5.3 Chi tiết integration

- Endpoint: `GET /api/business-partners/{businessPartnerId}/social-media/integrations/{integrationId}`
- Permission: `SOCIAL_MEDIA.FACEBOOK_INTEGRATION.VIEW`

Response 200:

```
{
  "id": "18181818-1818-1818-1818-181818181818",
  "businessPartnerId": "aaaaaaaa-aaaa-aaaa-aaaa-aaaaaaaaaaaa",
  "provider": {
    "id": "17171717-1717-1717-1717-171717171717",
    "name": "Facebook",
    "code": "FACEBOOK"
  },
  "appld": "123456789",
  "status": "Authorized",
  "authorizedAt": "2026-06-16T10:00:00Z",
  "lastSyncedAt": "2026-06-16T10:05:00Z",
  "externalUserId": "meta-user-id",
  "externalUserName": "Meta User",
  "pages": [
    {
      "id": "19191919-1919-1919-1919-191919191919",
      "externalPagelId": "page-id",
      "pageName": "SAR Coffee Page",
    }
  ]
}
```

```
"pageAvatarUrl": null,  
"isBotEnabled": true,  
"isActive": true,  
"schedules": []  
}  
]  
}
```

5.4 Tạo Facebook app config

- Endpoint: **POST**
/api/business-partners/{businessPartnerId}/social-media/facebook/app-config
- Permission: **SOCIAL_MEDIA.FACEBOOK_INTEGRATION.CREATE**

Request:

```
{  
  "appId": "123456789",  
  "appSecret": "meta-app-secret"  
}
```

Response:

```
{  
  "integrationId": "18181818-1818-1818-1818-181818181818",  
  "businessPartnerId": "aaaaaaaa-aaaa-aaaa-aaaa-aaaaaaaaaaaa",  
  "providerCode": "FACEBOOK",  
  "appId": "123456789",  
  "status": "Configured"  
}
```

FE note: không log/hiện lại appSecret sau khi submit.

5.5 Cập nhật Facebook app config

- Endpoint: **PUT**
/api/business-partners/{businessPartnerId}/social-media/facebook/app-config
- Permission: **SOCIAL_MEDIA.FACEBOOK_INTEGRATION.UPDATE**
- Body: giống create.
- Response: **FacebookAppConfigResponse**.

5.6 Bắt đầu OAuth Facebook

- Endpoint: **POST**
/api/business-partners/{businessPartnerId}/social-media/facebook/oauth/start

- Permission: **SOCIAL_MEDIA.FACEBOOK_INTEGRATION.REAUTHORIZE**

Request:

```
{
  "redirectUri": "https://app.example.com/facebook/oauth/callback"
}
```

Response:

```
{
  "authorizationUrl": "https://www.facebook.com/...",
  "state": "opaque-state"
}
```

FE note: redirect browser sang **authorizationUrl**.

5.7 OAuth callback Facebook

- Endpoint: **GET /api/social-media/facebook/oauth/callback?code={code}&state={state}**
- Auth: anonymous.
- Khi gọi: Meta redirect về backend callback trong flow Business Partner social-media.

Response:

```
{
  "success": true,
  "businessPartnerId": "aaaaaaaa-aaaa-aaaa-aaaa-aaaaaaaaaaaa",
  "integrationId": "18181818-1818-1818-1818-181818181818",
  "status": "Authorized",
  "message": "Facebook authorization completed."
}
```

Lỗi thường gặp: Facebook OAuth failed: ..., code is required., state is required., OAuth state is invalid., OAuth state is expired or already used.

5.8 Lấy Facebook pages

- Endpoint: **GET /api/business-partners/{businessPartnerId}/social-media/facebook/pages**
- Permission: **SOCIAL_MEDIA.FACEBOOK_INTEGRATION.VIEW**

Response:

```
{
  "integrationId": "18181818-1818-1818-1818-181818181818",
  "pages": [
```

```
{
  "externalPagelId": "page-id",
  "pageName": "SAR Coffee Page",
  "pageAvatarUrl": null,
  "isSelected": true,
  "isBotEnabled": true
}
```

5.9 Lưu/chọn Facebook pages

- Endpoint: **POST /api/business-partners/{businessPartnerId}/social-media/facebook/pages**
- Permission: **SOCIAL_MEDIA.FACEBOOK_INTEGRATION.UPDATE**

Request:

```
{
  "pages": [
    {
      "externalPagelId": "page-id",
      "pageName": "SAR Coffee Page",
      "pageAvatarUrl": null,
      "pageAccessToken": "page-access-token-from-meta",
      "schedules": [
        {
          "dayOfWeek": "Monday",
          "startTime": "08:00",
          "endTime": "17:30"
        }
      ]
    }
  ]
}
```

Response:

```
{
  "businessPartnerId": "aaaaaaaa-aaaa-aaaa-aaaa-aaaaaaaaaaaaa",
  "integrationId": "18181818-1818-1818-1818-181818181818",
  "createdOrUpdatedPages": [
    {
      "id": "19191919-1919-1919-1919-191919191919",
      "externalPagelId": "page-id",
      "pageName": "SAR Coffee Page",
      "isBotEnabled": true,
      "isActive": true,
      "schedulesCount": 1
    }
  ]
}
```

```
}  
],  
"message": "Facebook pages saved."  
}
```

Lỗi: At least one Facebook page is required., duplicate page, ExternalPageId is required., PageName is required., integration chưa authorized.

FE note: pageAccessToken là secret. Không lưu localStorage, không log console.

5.10 Bật/tắt bot cho page

- Endpoint: **PUT**
`/api/business-partners/{businessPartnerId}/social-media/pages/{pageId}/bot-status`
- Permission: **SOCIAL_MEDIA.FACEBOOK_INTEGRATION.UPDATE**

Request:

```
{  
  "isBotEnabled": true  
}
```

Response:

```
{  
  "pageId": "19191919-1919-1919-1919-191919191919",  
  "isBotEnabled": true  
}
```

5.11 Lấy lịch bot

- Endpoint: **GET**
`/api/business-partners/{businessPartnerId}/social-media/pages/{pageId}/schedules`
- Permission: **SOCIAL_MEDIA.BOT_SCHEDULE.VIEW**

Response:

```
[  
  {  
    "id": "20202020-2020-2020-2020-202020202020",  
    "dayOfWeek": "Monday",  
    "startTime": "08:00",  
    "endTime": "17:30",  
    "isActive": true  
  }  
]
```

5.12 Replace lịch bot

- Endpoint: **PUT**
`/api/business-partners/{businessPartnerId}/social-media/pages/{pageId}/schedules`
- Permission: **SOCIAL_MEDIA.BOT_SCHEDULE.UPDATE**

Request:

```
{
  "schedules": [
    {
      "dayOfWeek": "Monday",
      "startTime": "08:00",
      "endTime": "17:30"
    },
    {
      "dayOfWeek": "Friday",
      "startTime": "22:00",
      "endTime": "02:00"
    }
  ]
}
```

Response: list **BotWorkingScheduleResponse**.

Validate:

- dayOfWeek parse theo enum .NET DayOfWeek, ví dụ Monday, Tuesday.
- startTime, endTime đúng format HH:mm.
- Cho phép cả khung giờ qua ngày, ví dụ 22:00 đến 02:00.
- Không cho trùng/overlap sau khi expand qua ngày.

Lỗi: **DayOfWeek is invalid., StartTime must be in HH:mm format., EndTime must be in HH:mm format., StartTime and EndTime cannot be the same., Bot working schedules must not overlap.**

6. Meta OAuth client-driven integration

Đây là flow khác với mục 5. Flow này dùng endpoint `/api/integrations/meta`, FE lấy login URL, user authorize trên Meta, Meta redirect về FE, sau đó FE post `code/state/appSecret` về BE. BE đổi token, verify scopes, lấy pages và lưu page tokens dạng mã hóa. API response chỉ trả `hasAccessToken`, không trả raw token.

6.1 Tạo Meta login URL

- Endpoint: **GET**
`/api/integrations/meta/login-url?appId={appId}&redirectUrl={redirectUrl}&forceRerequest=false`

- Permission: **SOCIAL_MEDIA.FACEBOOK_INTEGRATION.REAUTHORIZE**

Response:

```
{
  "loginUrl": "https://www.facebook.com/v23.0/dialog/oauth?...",
  "state": "opaque-state",
  "expiresAt": "2026-06-16T10:10:00Z"
}
```

Validate:

- appId bắt buộc.
- redirectUrl bắt buộc và phải là absolute HTTPS URL.
- forceRerequest=true nếu FE muốn ép user cấp lại permission.

FE note: redirect user sang loginUrl. Khi Meta redirect về redirectUrl, FE đọc code, state, hoặc error trên query và gọi API callback bên dưới.

6.2 Xử lý Meta OAuth callback

- Endpoint: **POST /api/integrations/meta/oauth/callback**
- Auth: source hiện tại không gắn [Authorize].

Request thành công:

```
{
  "appId": "123456789",
  "appSecret": "meta-app-secret",
  "redirectUrl": "https://app.example.com/meta/callback",
  "code": "code-from-meta",
  "state": "state-from-login-url",
  "error": null,
  "errorReason": null,
  "errorDescription": null
}
```

Response **200**:

```
{
  "success": true,
  "message": "Meta OAuth completed successfully.",
  "pages": [
    {
      "pageId": "page-id",
      "pageName": "SAR Coffee Page",
      "avatarUrl": "https://...",
    }
  ]
}
```

```
    "hasAccessToken": true
  }
]
```

Request khi Meta trả lời error:

```
{
  "appId": "123456789",
  "appSecret": "meta-app-secret",
  "redirectUrl": "https://app.example.com/meta/callback",
  "code": null,
  "state": "state-from-login-url",
  "error": "access_denied",
  "errorReason": "user_denied",
  "errorDescription": "User denied access"
}
```

Response 400 riêng của controller:

```
{
  "success": false,
  "message": "User denied access",
  "pages": []
}
```

Lỗi validate/Meta:

- AppId is required.
- AppSecret is required.
- RedirectUrl is required.
- Meta OAuth callback code is required.
- Meta OAuth callback state is required.
- Meta OAuth state is invalid or expired.
- Không có page/token hợp lệ từ Meta.
- Token thiếu scope bắt buộc.
- Lỗi Graph API trả 502.

FE note bảo mật: appSecret và page access token là secret. Nếu có thể, nên để appSecret được lưu/cấu hình backend thay vì FE nắm giữ. Nếu FE vẫn phải submit theo source hiện tại, không log, không persist vào browser storage.

7. Webhooks

Webhook endpoints không dùng Bearer token vì Meta/Facebook gọi trực tiếp vào backend.

7.1 Verify Meta webhook

- Endpoint: **GET**
`/api/webhooks/meta/{channel}?hub.mode=subscribe&hub.verify_token={token}&hub.challenge={challenge}`
- Auth: anonymous.
- Khi gọi: Facebook Developer webhook verification.

Response 200: challenge từ query.

Ví dụ:

GET

`/api/webhooks/meta/facebook?hub.mode=subscribe&hub.verify_token=verify-token&hub.challenge=12345`

Lỗi hiện tại:

- Nếu hub.mode thiếu: 500 do provider ném `InvalidOperationException`.
- Nếu hub.mode khác subscribe: 500 với message `unsupported` trong `Development`.
- Source hiện tại có TODO compare hub.verify_token với token cấu hình/business.

FE note: endpoint này phục vụ Meta, không cần gọi từ UI ngoài trừ màn hình diagnostic/nội bộ.

7.2 Receive Meta webhook

- Endpoint: **POST** `/api/webhooks/meta/{provider}`
- Auth: anonymous.
- Header bắt buộc: `x-hub-signature-256`.
- Body: raw Meta webhook payload.

Request mẫu:

POST `/api/webhooks/meta/facebook`

`x-hub-signature-256: sha256=...`

`Content-Type: application/json`

```
{  
  "object": "page",
```

```

"entry": [
  {
    "id": "page-id",
    "time": 1710000000,
    "messaging": [
      {
        "sender": { "id": "psid" },
        "recipient": { "id": "page-id" },
        "timestamp": 1710000000,
        "message": {
          "mid": "message-id",
          "text": "Hello"
        }
      }
    ]
  }
]
}

```

Response thành công: 200 OK body rỗng.

Lỗi:

- 400 plain text: **Missing Meta webhook signature.**
- 401 plain text: **Invalid Meta webhook signature.**
- 400 plain text: **Webhook payload is required.**
- 400/500 JSON ErrorResponse cho lỗi mapper/provider khác.

Backend note: handler map payload thành inbound events, skip duplicate theo EventId, persist DB, rồi publish queue để xử lý async. Conversation automation được scope theo page/sender trong processing layer.

8. Tổng hợp endpoint nhanh

Meth od	Endpoint	Au th	Permission
POS T	/api/auth/login	No	-
POS T	/api/auth/refresh-token	No	-

POST	/api/auth/logout	No	-
GET	/api/me	Yes	-
GET	/api/me/permissions	Yes	-
GET	/api/business-partners	Yes	BUSINESS_PARTNER.PROFILE.VIEW
GET	/api/business-partners/{id}	Yes	BUSINESS_PARTNER.PROFILE.VIEW
POST	/api/business-partners	Yes	BUSINESS_PARTNER.PROFILE.CREATE
PUT	/api/business-partners/{id}	Yes	BUSINESS_PARTNER.PROFILE.UPDATE
DELETE	/api/business-partners/{id}	Yes	BUSINESS_PARTNER.PROFILE.DELETE
GET	/api/roles	Yes	ROLE_MANAGEMENT.ROLE_LIST.VIEW
GET	/api/roles/{id}	Yes	ROLE_MANAGEMENT.ROLE_LIST.VIEW
POST	/api/roles	Yes	ROLE_MANAGEMENT.ROLE_LIST.CREATE
PUT	/api/roles/{id}	Yes	ROLE_MANAGEMENT.ROLE_LIST.UPDATE
DELETE	/api/roles/{id}	Yes	ROLE_MANAGEMENT.ROLE_LIST.DELETE
GET	/api/roles/{id}/permissions/matrix	Yes	PERMISSION_MANAGEMENT.ROLE_PERMISSION_CONFIG.VIEW

PUT	/api/roles/{id}/permissions	Yes	PERMISSION_MANAGEMENT.ROLE_PERMISSION_CONFIG.UPDATE
PATCH	/api/roles/{id}/permissions	Yes	PERMISSION_MANAGEMENT.ROLE_PERMISSION_CONFIG.UPDATE
POST	/api/roles/{targetId}/permissions/copy-from/{sourceId}	Yes	PERMISSION_MANAGEMENT.ROLE_PERMISSION_CONFIG.UPDATE
GET	/api/features	Yes	PERMISSION_MANAGEMENT.FEATURE_CONFIG.VIEW
GET	/api/features/{id}	Yes	PERMISSION_MANAGEMENT.FEATURE_CONFIG.VIEW
POST	/api/features	Yes	PERMISSION_MANAGEMENT.FEATURE_CONFIG.CREATE
PUT	/api/features/{id}	Yes	PERMISSION_MANAGEMENT.FEATURE_CONFIG.UPDATE
DELETE	/api/features/{id}	Yes	PERMISSION_MANAGEMENT.FEATURE_CONFIG.DELETE
GET	/api/pages	Yes	PERMISSION_MANAGEMENT.PAGE_CONFIG.VIEW
GET	/api/pages/{id}	Yes	PERMISSION_MANAGEMENT.PAGE_CONFIG.VIEW
POST	/api/pages	Yes	PERMISSION_MANAGEMENT.PAGE_CONFIG.CREATE
PUT	/api/pages/{id}	Yes	PERMISSION_MANAGEMENT.PAGE_CONFIG.UPDATE
DELETE	/api/pages/{id}	Yes	PERMISSION_MANAGEMENT.PAGE_CONFIG.DELETE
GET	/api/permission-actions	Yes	PERMISSION_MANAGEMENT.ACTION_CONFIG.VIEW

POST	/api/permission-actions	Yes	PERMISSION_MANAGEMENT.ACTION_CONFIG.CREATE
PUT	/api/permission-actions/{id}	Yes	PERMISSION_MANAGEMENT.ACTION_CONFIG.UPDATE
DELETE	/api/permission-actions/{id}	Yes	PERMISSION_MANAGEMENT.ACTION_CONFIG.DELETE
GET	/api/page-actions	Yes	PERMISSION_MANAGEMENT.PAGE_ACTION_CONFIG.VIEW
POST	/api/page-actions	Yes	PERMISSION_MANAGEMENT.PAGE_ACTION_CONFIG.CREATE
PUT	/api/page-actions/{id}	Yes	PERMISSION_MANAGEMENT.PAGE_ACTION_CONFIG.UPDATE
DELETE	/api/page-actions/{id}	Yes	PERMISSION_MANAGEMENT.PAGE_ACTION_CONFIG.DELETE
GET	/api/system-module-definitions	Yes	PERMISSION_MANAGEMENT.FEATURE_CONFIG.VIEW
GET	/api/system-module-definitions/{id}/page-definitions	Yes	PERMISSION_MANAGEMENT.PAGE_CONFIG.VIEW
GET	/api/system-action-definitions	Yes	PERMISSION_MANAGEMENT.ACTION_CONFIG.VIEW
GET	/api/social-media/providers	Yes	SOCIAL_MEDIA.FACEBOOK_INTEGRATION.VIEW
GET	/api/business-partners/{id}/social-media/integrations	Yes	SOCIAL_MEDIA.FACEBOOK_INTEGRATION.VIEW
GET	/api/business-partners/{id}/social-media/integrations/{integrationId}	Yes	SOCIAL_MEDIA.FACEBOOK_INTEGRATION.VIEW
POST	/api/business-partners/{id}/social-media/facebook/app-config	Yes	SOCIAL_MEDIA.FACEBOOK_INTEGRATION.CREATE

PUT	/api/business-partners/{id}/social-media/facebook/app-config	Yes	SOCIAL_MEDIA.FACEBOOK_INTEGRATION.UPDATE
POST	/api/business-partners/{id}/social-media/facebook/oauth/start	Yes	SOCIAL_MEDIA.FACEBOOK_INTEGRATION.REAUTHORIZE
GET	/api/social-media/facebook/oauth/callback	No	-
GET	/api/business-partners/{id}/social-media/facebook/pages	Yes	SOCIAL_MEDIA.FACEBOOK_INTEGRATION.VIEW
POST	/api/business-partners/{id}/social-media/facebook/pages	Yes	SOCIAL_MEDIA.FACEBOOK_INTEGRATION.UPDATE
PUT	/api/business-partners/{id}/social-media/pages/{pageId}/bot-status	Yes	SOCIAL_MEDIA.FACEBOOK_INTEGRATION.UPDATE
GET	/api/business-partners/{id}/social-media/pages/{pageId}/schedules	Yes	SOCIAL_MEDIA.BOT_SCHEDULE.VIEW
PUT	/api/business-partners/{id}/social-media/pages/{pageId}/schedules	Yes	SOCIAL_MEDIA.BOT_SCHEDULE.UPDATE
GET	/api/integrations/meta/login-url	Yes	SOCIAL_MEDIA.FACEBOOK_INTEGRATION.REAUTHORIZE
POST	/api/integrations/meta/oauth/callback	No	-
GET	/api/webhooks/meta/{channel}	No	-
POST	/api/webhooks/meta/{provider}	No	-

9. Ghi chú vận hành và FE integration

- Dùng traceId trong ErrorResponse để đối chiếu log backend.
- Không đưa raw secret vào log FE: appSecret, pageAccessToken, refresh token.

- Meta tokens được backend mã hóa; khi deploy container/multiple instances cần key-ring Data Protection persistent/shared, nếu không token đã mã hóa có thể không decrypt được sau restart/redeploy.
- Runtime migration chỉ nên bật khi có chủ đích qua `AUTO_MIGRATE_DATABASE=true`.
- SMTP cần cấu hình đúng nếu muốn business partner create flow gửi email thành công.
- Endpoint webhook POST cần raw body đúng nguyên để validate signature; proxy/gateway không nên mutate body.